Aim:

1.Understanding the working of NIC cards, Ethernet/Fast Ethernet/Gigabit Ethernet.

i)NIC Cards:

Working:

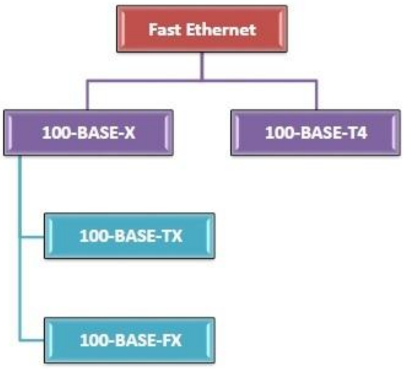
1.Converts data from the computer into signals that can be sent over the network.
2.Uses a unique MAC address to identify the device.
3.Transfers data frames to and from the network medium.
4.Handles error detection and communication control at the data link layer.

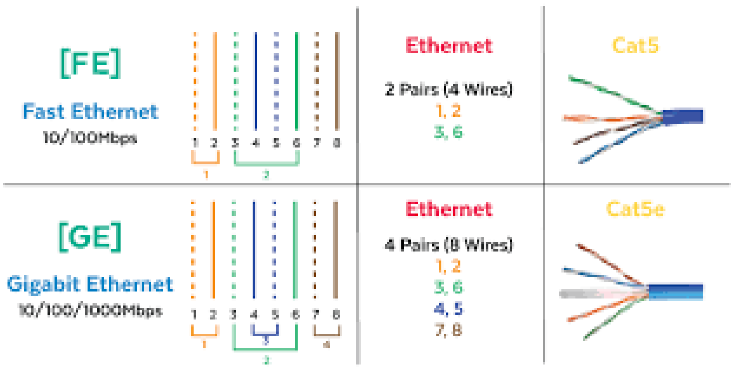
ii)Ethernet:

Working:

1.A widely used LAN technology that sends data in the form of frames.
2.Uses CSMA/CD for managing access to the shared medium.
3.Supports speeds like 10 Mbps.
4.Uses twisted pair, coaxial or fiber cables.

iii)Fast Ethernet:





1.An improved version of Ethernet offering speeds up to 100 Mbps.
2.Maintains the same frame format as Ethernet for compatibility.
3.Uses twisted pair or fiber cables.
4.Designed for higher speed LAN communication.

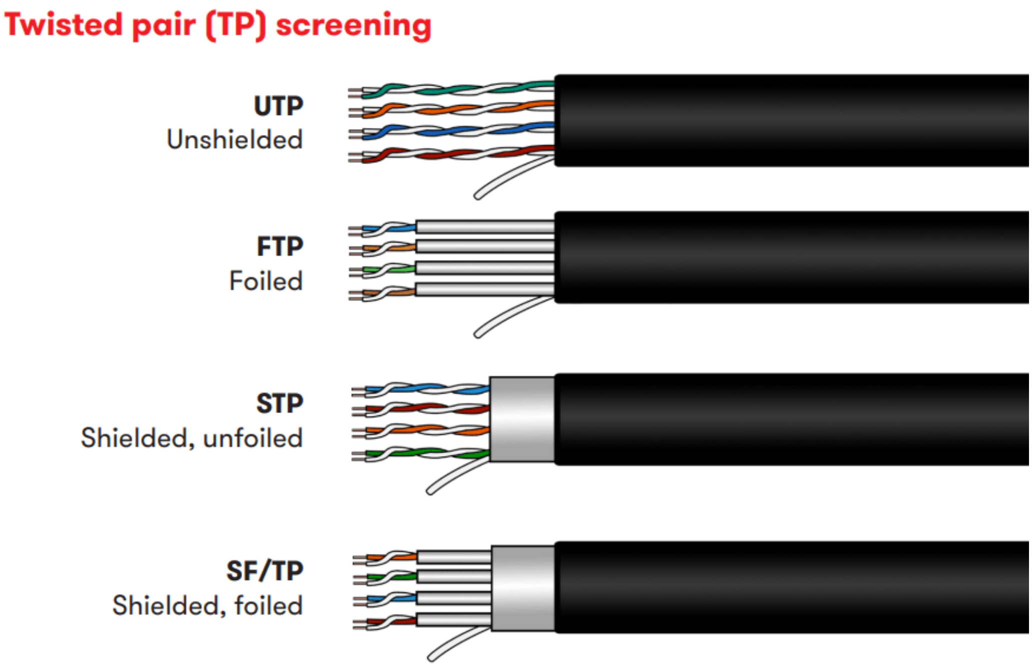
iv)Gigabit Ethernet:

1.Provides data speeds of 1 Gbps (1000 Mbps).
2.Used in high-performance networks needing fast data transfer.
3.Works on copper (Cat5e/Cat6) or fiber optic cables.
4.Supports full-duplex communication for better efficiency.

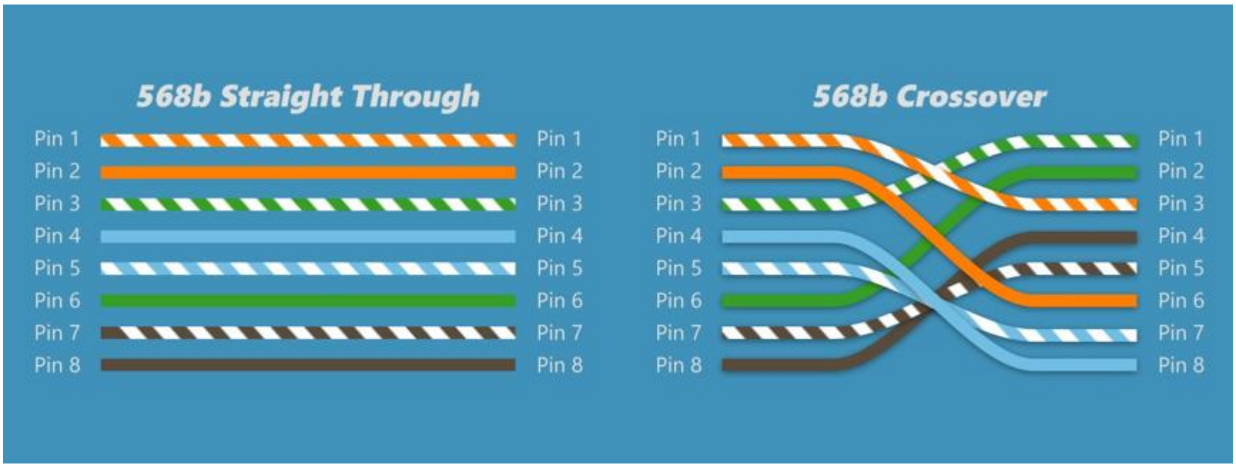
2.Crimping of Twisted-Pair Cable with RJ45connector for Straight- through, Cross-Over, Roll-Over.

i)RJ45 LAN Port:

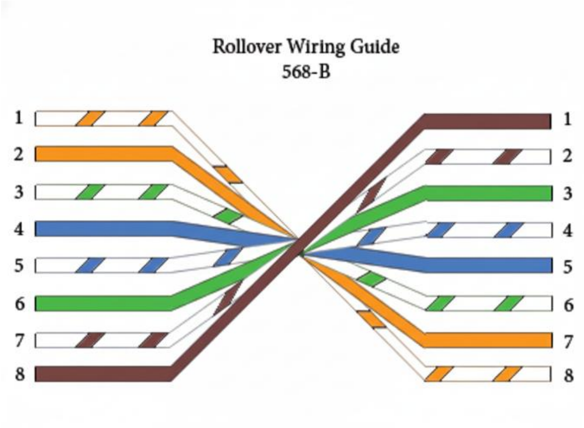
ii)Twisted pair cable:



iii)Straight Through, Cross Over



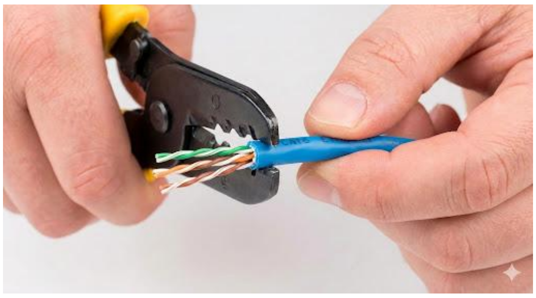
iv)Roll Over:



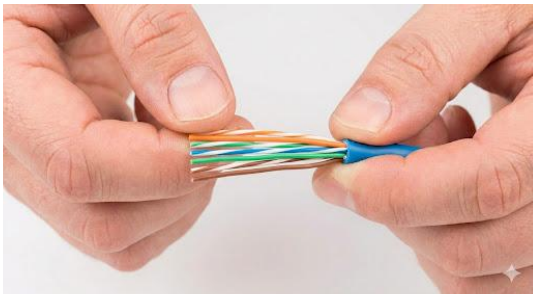
v)Crimping Tool:

Steps for Crimping of Twisted-Pair Cable with RJ45connector:

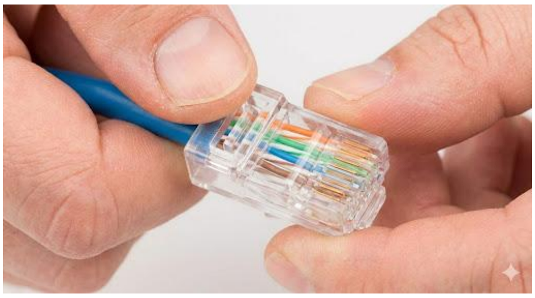
1.Strip the Cable Jacket: First, you'll need to remove a small section of the outer jacket of the Ethernet cable. Use a cable stripper to carefully cut around the jacket, being careful not to nick the wires inside.



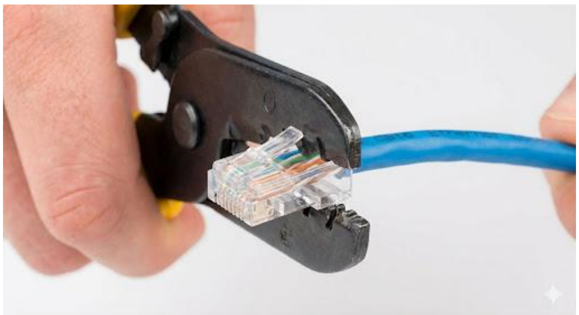
2.Untwist and Arrange the Wires: Once the jacket is removed, you'll see four twisted pairs of wires. Untwist them and arrange them according to the one of the standards. **Ex:** (Orange-White, Orange, Green-White, Blue, Blue-White, Green, Brown-White, Brown). Straighten them out and trim them so they are all the same length.



3.Insert the Wires into the Connector: Carefully insert the arranged wires into the RJ45 connector. Make sure each wire goes into its own channel and reaches the end of the connector. The outer jacket of the cable should also be pushed inside the connector for strain relief.



4.Crimp the Connector: Finally, place the connector with the wires inserted into your crimping tool. Squeeze the handles of the tool firmly. This action will push the metal contacts of the connector into the wires, creating a solid electrical connection, and also secure the cable jacket.



After crimping, it's always a good idea to test your new cable with a cable tester to ensure all the connections are correct.